

FagusED · Co-ordinated Sciences
B1 · Characteristics and classification of livi...

Define: species.

COS-0001

PREGUNTA

FagusED · Co-ordinated Sciences
B1 · Characteristics and classification of livi...

Define: binomial system.

COS-0002

PREGUNTA

FagusED · Co-ordinated Sciences
B1 · Characteristics and classification of livi...

Define: dichotomous key.

COS-0003

PREGUNTA

FagusED · Co-ordinated Sciences
B1 · Characteristics and classification of livi...

What is the main idea of characteristics of life?

COS-0004

PREGUNTA

FagusED · Co-ordinated Sciences
B2 · Cells

Define: tissue.

COS-0005

PREGUNTA

FagusED · Co-ordinated Sciences
B2 · Cells

Define: organ.

COS-0006

PREGUNTA

FagusED · Co-ordinated Sciences
B2 · Cells

Define: organelle.

COS-0007

PREGUNTA

FagusED · Co-ordinated Sciences
B2 · Cells

What is the main idea of cell structures?

COS-0008

PREGUNTA

FagusED · Co-ordinated Sciences
B3 · Movement into and out of cells

Define: diffusion.

COS-0009

PREGUNTA

FagusED · Co-ordinated Sciences
B3 · Movement into and out of cells

Define: osmosis.

COS-0010

PREGUNTA

FagusED · Co-ordinated Sciences
B3 · Movement into and out of cells

Define: active transport.

COS-0011

PREGUNTA

FagusED · Co-ordinated Sciences
B3 · Movement into and out of cells

What is the main idea of diffusion?

COS-0012

PREGUNTA

FagusED · Co-ordinated Sciences
B4 · Biological molecules

Define: reducing sugar.

COS-0013

PREGUNTA

FagusED · Co-ordinated Sciences
B4 · Biological molecules

Define: protein.

COS-0014

PREGUNTA

FagusED · Co-ordinated Sciences
B4 · Biological molecules

Define: lipid.

COS-0015

PREGUNTA

FagusED · Co-ordinated Sciences
B1 · Characteristics and classification of living organisms

a sequence of paired contrasting statements used for identification

COS-0003

RESPUESTA

FagusED · Co-ordinated Sciences
B1 · Characteristics and classification of living organisms

a two-part scientific name consisting of genus and species

COS-0002

RESPUESTA

FagusED · Co-ordinated Sciences
B1 · Characteristics and classification of living organisms

a group of organisms that can reproduce to produce fertile offspring

COS-0001

RESPUESTA

FagusED · Co-ordinated Sciences
B2 · Cells

a structure made of different tissues working together

COS-0006

RESPUESTA

FagusED · Co-ordinated Sciences
B2 · Cells

a group of cells with similar structures working together for a function

COS-0005

RESPUESTA

FagusED · Co-ordinated Sciences
B1 · Characteristics and classification of living organisms

Living organisms show movement, respiration, sensitivity, growth, reproduction, excretion and nutrition. These characteristics describe coordinated cellular processes; no single characteristic alone is always sufficient to decide whether an unfamiliar object is living.

COS-0004

RESPUESTA

FagusED · Co-ordinated Sciences
B3 · Movement into and out of cells

net movement from higher to lower concentration due to random motion

COS-0009

RESPUESTA

FagusED · Co-ordinated Sciences
B2 · Cells

The cell membrane controls movement of substances; cytoplasm is the site of many reactions; the nucleus contains genetic material; mitochondria carry out aerobic respiration; ribosomes synthesise proteins. Plant cells additionally have a cellulose wall, permanent vacuole and chloroplasts in photosynthetic tissues.

COS-0008

RESPUESTA

FagusED · Co-ordinated Sciences
B2 · Cells

a specialised structure within a cell

COS-0007

RESPUESTA

FagusED · Co-ordinated Sciences
B3 · Movement into and out of cells

Diffusion is the net movement of particles from higher to lower concentration due to random motion. A steeper concentration gradient, higher temperature and larger surface area increase rate; a shorter distance also increases exchange.

COS-0012

RESPUESTA

FagusED · Co-ordinated Sciences
B3 · Movement into and out of cells

movement against a concentration gradient using energy

COS-0011

RESPUESTA

FagusED · Co-ordinated Sciences
B3 · Movement into and out of cells

net movement of water from higher to lower water potential through a partially permeable membrane

COS-0010

RESPUESTA

FagusED · Co-ordinated Sciences
B4 · Biological molecules

a molecule formed from fatty acids and glycerol

COS-0015

RESPUESTA

FagusED · Co-ordinated Sciences
B4 · Biological molecules

a polymer of amino acids folded into a specific shape

COS-0014

RESPUESTA

FagusED · Co-ordinated Sciences
B4 · Biological molecules

a sugar that reduces Benedict's reagent when heated

COS-0013

RESPUESTA

FagusED · Co-ordinated Sciences
B4 · Biological molecules

What is the main idea of carbohydrates?

COS-0016

PREGUNTA

FagusED · Co-ordinated Sciences
B5 · Enzymes

Define: enzyme.

COS-0017

PREGUNTA

FagusED · Co-ordinated Sciences
B5 · Enzymes

Define: active site.

COS-0018

PREGUNTA

FagusED · Co-ordinated Sciences
B5 · Enzymes

Define: substrate.

COS-0019

PREGUNTA

FagusED · Co-ordinated Sciences
B5 · Enzymes

What is the main idea of enzyme action?

COS-0020

PREGUNTA

FagusED · Co-ordinated Sciences
B6 · Plant nutrition

Define: photosynthesis.

COS-0021

PREGUNTA

FagusED · Co-ordinated Sciences
B6 · Plant nutrition

Define: limiting factor.

COS-0022

PREGUNTA

FagusED · Co-ordinated Sciences
B6 · Plant nutrition

Define: stoma.

COS-0023

PREGUNTA

FagusED · Co-ordinated Sciences
B6 · Plant nutrition

What is the main idea of photosynthesis?

COS-0024

PREGUNTA

FagusED · Co-ordinated Sciences
B7 · Human nutrition

Define: ingestion.

COS-0025

PREGUNTA

FagusED · Co-ordinated Sciences
B7 · Human nutrition

Define: digestion.

COS-0026

PREGUNTA

FagusED · Co-ordinated Sciences
B7 · Human nutrition

Define: absorption.

COS-0027

PREGUNTA

FagusED · Co-ordinated Sciences
B7 · Human nutrition

What is the main idea of balanced diet?

COS-0028

PREGUNTA

FagusED · Co-ordinated Sciences
B8 · Transport in plants

Define: transpiration.

COS-0029

PREGUNTA

FagusED · Co-ordinated Sciences
B8 · Transport in plants

Define: translocation.

COS-0030

PREGUNTA

FagusED · Co-ordinated Sciences
B5 · Enzymes

the region of an enzyme where the substrate binds

COS-0018

RESPUESTA

FagusED · Co-ordinated Sciences
B5 · Enzymes

a protein that acts as a biological catalyst

COS-0017

RESPUESTA

FagusED · Co-ordinated Sciences
B4 · Biological molecules

Glucose is a simple reducing sugar used in respiration. Starch and glycogen are large insoluble storage carbohydrates made from glucose; cellulose forms plant cell walls. Insolubility prevents storage molecules affecting water potential and allows compact storage.

COS-0016

RESPUESTA

FagusED · Co-ordinated Sciences
B6 · Plant nutrition

the process that makes carbohydrates from carbon dioxide and water using light energy

COS-0021

RESPUESTA

FagusED · Co-ordinated Sciences
B5 · Enzymes

Enzymes are proteins that catalyse metabolic reactions. A substrate with a complementary shape binds at the active site to form an enzyme-substrate complex; products leave and the enzyme can be reused. Enzymes lower activation energy.

COS-0020

RESPUESTA

FagusED · Co-ordinated Sciences
B5 · Enzymes

the reactant molecule on which an enzyme acts

COS-0019

RESPUESTA

FagusED · Co-ordinated Sciences
B6 · Plant nutrition

Photosynthesis transfers light energy into chemical energy by producing glucose from carbon dioxide and water; oxygen is released. Chlorophyll absorbs light. Glucose may be respired, converted to starch or cellulose, combined with nitrate to make amino acids, or transported as sucrose.

COS-0024

RESPUESTA

FagusED · Co-ordinated Sciences
B6 · Plant nutrition

a pore in the leaf epidermis controlled by guard cells

COS-0023

RESPUESTA

FagusED · Co-ordinated Sciences
B6 · Plant nutrition

a factor in shortest supply that restricts reaction rate

COS-0022

RESPUESTA

FagusED · Co-ordinated Sciences
B7 · Human nutrition

movement of digested molecules through the intestinal wall into blood or lymph

COS-0027

RESPUESTA

FagusED · Co-ordinated Sciences
B7 · Human nutrition

breakdown of large insoluble molecules into small soluble molecules

COS-0026

RESPUESTA

FagusED · Co-ordinated Sciences
B7 · Human nutrition

taking substances such as food and drink into the body through the mouth

COS-0025

RESPUESTA

FagusED · Co-ordinated Sciences
B8 · Transport in plants

movement of sucrose and amino acids in phloem

COS-0030

RESPUESTA

FagusED · Co-ordinated Sciences
B8 · Transport in plants

loss of water vapour from plant leaves

COS-0029

RESPUESTA

FagusED · Co-ordinated Sciences
B7 · Human nutrition

A balanced diet supplies carbohydrate, protein, lipid, vitamins, minerals, fibre and water in proportions matched to age, sex and activity. Energy imbalance can cause obesity or mass loss. Vitamin C prevents scurvy; vitamin D and calcium support bones; iron is needed for haemoglobin.

COS-0028

RESPUESTA

FagusED · Co-ordinated Sciences
B8 · Transport in plants

Define: xylem.

COS-0031

PREGUNTA

FagusED · Co-ordinated Sciences
B8 · Transport in plants

What is the main idea of xylem?

COS-0032

PREGUNTA

FagusED · Co-ordinated Sciences
B9 · Transport in animals

Define: artery.

COS-0033

PREGUNTA

FagusED · Co-ordinated Sciences
B9 · Transport in animals

Define: vein.

COS-0034

PREGUNTA

FagusED · Co-ordinated Sciences
B9 · Transport in animals

Define: capillary.

COS-0035

PREGUNTA

FagusED · Co-ordinated Sciences
B9 · Transport in animals

What is the main idea of double circulation?

COS-0036

PREGUNTA

FagusED · Co-ordinated Sciences
B10 · Diseases and immunity

Define: pathogen.

COS-0037

PREGUNTA

FagusED · Co-ordinated Sciences
B10 · Diseases and immunity

Define: antigen.

COS-0038

PREGUNTA

FagusED · Co-ordinated Sciences
B10 · Diseases and immunity

Define: antibody.

COS-0039

PREGUNTA

FagusED · Co-ordinated Sciences
B10 · Diseases and immunity

What is the main idea of pathogens and transmission?

COS-0040

PREGUNTA

FagusED · Co-ordinated Sciences
B11 · Gas exchange in humans

Define: ventilation.

COS-0041

PREGUNTA

FagusED · Co-ordinated Sciences
B11 · Gas exchange in humans

Define: gas exchange.

COS-0042

PREGUNTA

FagusED · Co-ordinated Sciences
B11 · Gas exchange in humans

Define: alveolus.

COS-0043

PREGUNTA

FagusED · Co-ordinated Sciences
B11 · Gas exchange in humans

What is the main idea of gas exchange system?

COS-0044

PREGUNTA

FagusED · Co-ordinated Sciences
B12 · Respiration

Define: respiration.

COS-0045

PREGUNTA

FagusED · Co-ordinated Sciences
B9 · Transport in animals

a blood vessel carrying blood away from the heart

COS-0033

RESPUESTA

FagusED · Co-ordinated Sciences
B8 · Transport in plants

Xylem vessels are dead, hollow tubes with lignified walls. They carry water and mineral ions upward and also support the plant. No end walls and a continuous lumen reduce resistance; lignin prevents collapse under tension.

COS-0032

RESPUESTA

FagusED · Co-ordinated Sciences
B8 · Transport in plants

tissue transporting water and mineral ions and providing support

COS-0031

RESPUESTA

FagusED · Co-ordinated Sciences
B9 · Transport in animals

Humans have a closed double circulation: heart to lungs and back, then heart to body and back. Separation maintains low pulmonary pressure to protect lungs and high systemic pressure for rapid delivery. Fish have a single circulation.

COS-0036

RESPUESTA

FagusED · Co-ordinated Sciences
B9 · Transport in animals

a microscopic vessel with a one-cell-thick wall for exchange

COS-0035

RESPUESTA

FagusED · Co-ordinated Sciences
B9 · Transport in animals

a vessel carrying blood toward the heart

COS-0034

RESPUESTA

FagusED · Co-ordinated Sciences
B10 · Diseases and immunity

a specific protein produced by lymphocytes that binds an antigen

COS-0039

RESPUESTA

FagusED · Co-ordinated Sciences
B10 · Diseases and immunity

a molecule recognised as foreign that stimulates an immune response

COS-0038

RESPUESTA

FagusED · Co-ordinated Sciences
B10 · Diseases and immunity

an organism that causes disease

COS-0037

RESPUESTA

FagusED · Co-ordinated Sciences
B11 · Gas exchange in humans

diffusion of oxygen and carbon dioxide between alveoli and blood

COS-0042

RESPUESTA

FagusED · Co-ordinated Sciences
B11 · Gas exchange in humans

movement of air into and out of the lungs

COS-0041

RESPUESTA

FagusED · Co-ordinated Sciences
B10 · Diseases and immunity

A pathogen is a disease-causing organism and may be a bacterium, virus, fungus or protist. Transmission can be direct, by droplets or body fluids, or indirect through food, water, surfaces or vectors. Good sanitation, safe water, hygiene and vector control break transmission routes.

COS-0040

RESPUESTA

FagusED · Co-ordinated Sciences
B12 · Respiration

chemical reactions in cells that break down nutrient molecules and release energy

COS-0045

RESPUESTA

FagusED · Co-ordinated Sciences
B11 · Gas exchange in humans

Air passes through trachea, bronchi and bronchioles to alveoli. Cartilage keeps larger airways open; mucus traps particles and cilia move it away. Gas exchange is diffusion of oxygen into blood and carbon dioxide out.

COS-0044

RESPUESTA

FagusED · Co-ordinated Sciences
B11 · Gas exchange in humans

a microscopic air sac where gas exchange occurs

COS-0043

RESPUESTA

FagusED · Co-ordinated Sciences
B12 · Respiration

Define: aerobic respiration.

COS-0046

PREGUNTA

FagusED · Co-ordinated Sciences
B12 · Respiration

Define: anaerobic respiration.

COS-0047

PREGUNTA

FagusED · Co-ordinated Sciences
B12 · Respiration

What is the main idea of aerobic respiration?

COS-0048

PREGUNTA

FagusED · Co-ordinated Sciences
B13 · Coordination and response

Define: stimulus.

COS-0049

PREGUNTA

FagusED · Co-ordinated Sciences
B13 · Coordination and response

Define: receptor.

COS-0050

PREGUNTA

FagusED · Co-ordinated Sciences
B13 · Coordination and response

Define: effector.

COS-0051

PREGUNTA

FagusED · Co-ordinated Sciences
B13 · Coordination and response

What is the main idea of nervous system?

COS-0052

PREGUNTA

FagusED · Co-ordinated Sciences
B14 · Drugs

Define: drug.

COS-0053

PREGUNTA

FagusED · Co-ordinated Sciences
B14 · Drugs

Define: antibiotic.

COS-0054

PREGUNTA

FagusED · Co-ordinated Sciences
B14 · Drugs

Define: tolerance.

COS-0055

PREGUNTA

FagusED · Co-ordinated Sciences
B14 · Drugs

What is the main idea of medicinal drugs?

COS-0056

PREGUNTA

FagusED · Co-ordinated Sciences
B15 · Reproduction

Define: pollination.

COS-0057

PREGUNTA

FagusED · Co-ordinated Sciences
B15 · Reproduction

Define: fertilisation.

COS-0058

PREGUNTA

FagusED · Co-ordinated Sciences
B15 · Reproduction

Define: implantation.

COS-0059

PREGUNTA

FagusED · Co-ordinated Sciences
B15 · Reproduction

What is the main idea of asexual and sexual reproduction?

COS-0060

PREGUNTA

FagusED · Co-ordinated Sciences
B12 · Respiration

Aerobic respiration is the chemical reactions in cells that use oxygen to break down nutrient molecules and release energy. Glucose and oxygen form carbon dioxide and water. Energy supports muscle contraction, active transport, protein synthesis, cell division and temperature maintenance.

COS-0048

RESPUESTA

FagusED · Co-ordinated Sciences
B12 · Respiration

energy release without oxygen

COS-0047

RESPUESTA

FagusED · Co-ordinated Sciences
B12 · Respiration

energy-releasing breakdown of nutrients using oxygen

COS-0046

RESPUESTA

FagusED · Co-ordinated Sciences
B13 · Coordination and response

a muscle or gland that carries out a response

COS-0051

RESPUESTA

FagusED · Co-ordinated Sciences
B13 · Coordination and response

a cell or sense organ detecting a stimulus

COS-0050

RESPUESTA

FagusED · Co-ordinated Sciences
B13 · Coordination and response

a change in the internal or external environment detected by a receptor

COS-0049

RESPUESTA

FagusED · Co-ordinated Sciences
B14 · Drugs

a medicine that kills bacteria or inhibits bacterial growth

COS-0054

RESPUESTA

FagusED · Co-ordinated Sciences
B14 · Drugs

a substance taken into the body that modifies chemical reactions

COS-0053

RESPUESTA

FagusED · Co-ordinated Sciences
B13 · Coordination and response

Receptors detect stimuli; impulses travel along sensory neurones to the central nervous system, across synapses using neurotransmitters, then along motor neurones to effectors. Reflexes are rapid automatic responses that protect the body.

COS-0052

RESPUESTA

FagusED · Co-ordinated Sciences
B15 · Reproduction

transfer of pollen from anther to stigma

COS-0057

RESPUESTA

FagusED · Co-ordinated Sciences
B14 · Drugs

A drug is a substance taken into the body that modifies chemical reactions. Medicinal drugs prevent or treat disease or relieve symptoms and should be tested for effectiveness, safe dose and side effects. Placebo-controlled, double-blind trials reduce bias.

COS-0056

RESPUESTA

FagusED · Co-ordinated Sciences
B14 · Drugs

needing a larger dose for the same effect

COS-0055

RESPUESTA

FagusED · Co-ordinated Sciences
B15 · Reproduction

Asexual reproduction uses one parent, no gamete fusion and produces genetically identical offspring except for mutation. It is rapid and effective in stable conditions. Sexual reproduction fuses haploid gametes, produces variation and may improve adaptation when conditions change, but requires more time and energy.

COS-0060

RESPUESTA

FagusED · Co-ordinated Sciences
B15 · Reproduction

attachment of an early embryo to the uterine lining

COS-0059

RESPUESTA

FagusED · Co-ordinated Sciences
B15 · Reproduction

fusion of nuclei of male and female gametes

COS-0058

RESPUESTA

FagusED · Co-ordinated Sciences
B16 · Inheritance

Define: gene.

COS-0061

PREGUNTA

FagusED · Co-ordinated Sciences
B16 · Inheritance

Define: allele.

COS-0062

PREGUNTA

FagusED · Co-ordinated Sciences
B16 · Inheritance

Define: genotype.

COS-0063

PREGUNTA

FagusED · Co-ordinated Sciences
B16 · Inheritance

What is the main idea of genetic information?

COS-0064

PREGUNTA

FagusED · Co-ordinated Sciences
B17 · Variation and selection

Define: variation.

COS-0065

PREGUNTA

FagusED · Co-ordinated Sciences
B17 · Variation and selection

Define: mutation.

COS-0066

PREGUNTA

FagusED · Co-ordinated Sciences
B17 · Variation and selection

Define: adaptation.

COS-0067

PREGUNTA

FagusED · Co-ordinated Sciences
B17 · Variation and selection

What is the main idea of variation?

COS-0068

PREGUNTA

FagusED · Co-ordinated Sciences
B18 · Organisms and their environment

Define: population.

COS-0069

PREGUNTA

FagusED · Co-ordinated Sciences
B18 · Organisms and their environment

Define: community.

COS-0070

PREGUNTA

FagusED · Co-ordinated Sciences
B18 · Organisms and their environment

Define: ecosystem.

COS-0071

PREGUNTA

FagusED · Co-ordinated Sciences
B18 · Organisms and their environment

What is the main idea of energy flow?

COS-0072

PREGUNTA

FagusED · Co-ordinated Sciences
B19 · Human influences on ecosystems

Define: biodiversity.

COS-0073

PREGUNTA

FagusED · Co-ordinated Sciences
B19 · Human influences on ecosystems

Define: bioaccumulation.

COS-0074

PREGUNTA

FagusED · Co-ordinated Sciences
B19 · Human influences on ecosystems

Define: biomagnification.

COS-0075

PREGUNTA

FagusED · Co-ordinated Sciences
B16 · Inheritance

the alleles an organism possesses

COS-0063

RESPUESTA

FagusED · Co-ordinated Sciences
B16 · Inheritance

an alternative form of a gene

COS-0062

RESPUESTA

FagusED · Co-ordinated Sciences
B16 · Inheritance

a length of DNA that codes for a protein

COS-0061

RESPUESTA

FagusED · Co-ordinated Sciences
B17 · Variation and selection

a random change in genetic material

COS-0066

RESPUESTA

FagusED · Co-ordinated Sciences
B17 · Variation and selection

differences between individuals of the same species

COS-0065

RESPUESTA

FagusED · Co-ordinated Sciences
B16 · Inheritance

A chromosome is a DNA molecule carrying genes. A gene is a length of DNA coding for a protein; an allele is an alternative form of a gene. Base sequence determines amino-acid sequence and protein shape. Most body cells are diploid with homologous chromosome pairs.

COS-0064

RESPUESTA

FagusED · Co-ordinated Sciences
B18 · Organisms and their environment

all organisms of one species in an area at a given time

COS-0069

RESPUESTA

FagusED · Co-ordinated Sciences
B17 · Variation and selection

Continuous variation shows a range and is often controlled by many genes plus environment, such as height. Discontinuous variation has distinct categories and is often controlled by one or few genes, such as ABO blood group. Genetic variation arises from mutation, meiosis and fertilisation.

COS-0068

RESPUESTA

FagusED · Co-ordinated Sciences
B17 · Variation and selection

an inherited feature increasing fitness in an environment

COS-0067

RESPUESTA

FagusED · Co-ordinated Sciences
B18 · Organisms and their environment

The Sun is the main energy source. Producers transfer light energy into chemical energy; consumers obtain it by feeding. Energy is lost between trophic levels in respiration, movement, heat, waste and uneaten material, so food chains are short and biomass decreases upward.

COS-0072

RESPUESTA

FagusED · Co-ordinated Sciences
B18 · Organisms and their environment

a community and its interactions with the physical environment

COS-0071

RESPUESTA

FagusED · Co-ordinated Sciences
B18 · Organisms and their environment

all populations of different species in an area

COS-0070

RESPUESTA

FagusED · Co-ordinated Sciences
B19 · Human influences on ecosystems

increase in pollutant concentration at successive trophic levels

COS-0075

RESPUESTA

FagusED · Co-ordinated Sciences
B19 · Human influences on ecosystems

build-up of a substance within an organism over time

COS-0074

RESPUESTA

FagusED · Co-ordinated Sciences
B19 · Human influences on ecosystems

the variety of genes, species and ecosystems in an area

COS-0073

RESPUESTA

FagusED · Co-ordinated Sciences
B19 · Human influences on ecosystems

What is the main idea of food production?

COS-0076

PREGUNTA

FagusED · Co-ordinated Sciences
C1 · States of matter

Which statement describes particles in a liquid?
A) They vibrate only at fixed positions.
B) They are close and move past one another.
C) They are very far apart and stationary.
D) They form a regular lattice and cannot move.

COS-0077

PREGUNTA

FagusED · Co-ordinated Sciences
C1 · States of matter

Which process occurs throughout a liquid at a fixed temperature?
A) evaporation
B) sublimation
C) boiling
D) diffusion

COS-0078

PREGUNTA

FagusED · Co-ordinated Sciences
C1 · States of matter

Why does gas pressure increase when a sealed gas is heated?
A) Particles become larger.
B) Particles collide more often and harder.
C) The number of particles increases.
D) Attractive forces become stronger.

COS-0079

PREGUNTA

FagusED · Co-ordinated Sciences
C1 · States of matter

Explain the particle model.

COS-0080

PREGUNTA

FagusED · Co-ordinated Sciences
C2 · Atoms, elements and compounds

Which statement defines a compound?
A) One type of atom only
B) Elements chemically combined in fixed proportions
C) Substances mixed in any proportion
D) A substance separable by filtration

COS-0081

PREGUNTA

FagusED · Co-ordinated Sciences
C2 · Atoms, elements and compounds

An atom has proton number 12 and nucleon number 25. How many neutrons?
A) 12
B) 13
C) 25
D) 37

COS-0082

PREGUNTA

FagusED · Co-ordinated Sciences
C2 · Atoms, elements and compounds

Which particles determine an element's chemical properties?
A) nuclear neutrons
B) outer-shell electrons
C) all nucleons
D) only inner electrons

COS-0083

PREGUNTA

FagusED · Co-ordinated Sciences
C2 · Atoms, elements and compounds

Explain elements, compounds and mixtures.

COS-0084

PREGUNTA

FagusED · Co-ordinated Sciences
C3 · Stoichiometry

What is the M_r of CaCO_3 ? Ar: Ca 40, C 12, O 16.
A) 68
B) 84
C) 100
D) 116

COS-0085

PREGUNTA

FagusED · Co-ordinated Sciences
C3 · Stoichiometry

How many moles are in 11 g of CO_2 ?
A) 0.25
B) 0.50
C) 2.0
D) 4.0

COS-0086

PREGUNTA

FagusED · Co-ordinated Sciences
C3 · Stoichiometry

Which is a correct balanced equation?
A) $\text{H}_2 + \text{O}_2 \rightarrow \text{H}_2\text{O}$
B) $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$
C) $\text{H}_2 + 2\text{O}_2 \rightarrow \text{H}_2\text{O}$
D) $2\text{H}_2 + 2\text{O}_2 \rightarrow \text{H}_2\text{O}$

COS-0087

PREGUNTA

FagusED · Co-ordinated Sciences
C1 · States of matter

C) boiling — Boiling occurs throughout the liquid at its boiling point.

COS-0078

RESPUESTA

FagusED · Co-ordinated Sciences
C1 · States of matter

B) They are close and move past one another. — Liquid particles remain close but can change neighbours.

COS-0077

RESPUESTA

FagusED · Co-ordinated Sciences
B19 · Human influences on ecosystems

Agricultural yield can increase through fertilisers, irrigation, pesticides, selective breeding, machinery and controlled environments. Intensive systems improve output but may reduce biodiversity, pollute water, select resistant pests and affect animal welfare. Sustainable approaches balance yield with long-term soil and ecosystem health.

COS-0076

RESPUESTA

FagusED · Co-ordinated Sciences
C2 · Atoms, elements and compounds

B) Elements chemically combined in fixed proportions — A compound contains chemically bonded elements in fixed ratios.

COS-0081

RESPUESTA

FagusED · Co-ordinated Sciences
C1 · States of matter

All matter is made of particles in constant motion. In a solid, particles are closely packed in a regular arrangement and vibrate about fixed positions. In a liquid, particles remain close but move past one another. In a gas, particles are far apart, move rapidly and randomly, and collide with the container.

COS-0080

RESPUESTA

FagusED · Co-ordinated Sciences
C1 · States of matter

B) Particles collide more often and harder. — Higher kinetic energy produces more frequent, larger-force collisions.

COS-0079

RESPUESTA

FagusED · Co-ordinated Sciences
C2 · Atoms, elements and compounds

An element contains one type of atom. A compound contains two or more elements chemically combined in fixed proportions and can be separated only chemically. A mixture contains substances not chemically combined; its components retain their properties and can be separated physically.

COS-0084

RESPUESTA

FagusED · Co-ordinated Sciences
C2 · Atoms, elements and compounds

B) outer-shell electrons — Bonding and reactions involve valence electrons.

COS-0083

RESPUESTA

FagusED · Co-ordinated Sciences
C2 · Atoms, elements and compounds

B) $13 - 25 - 12 = 13$.

COS-0082

RESPUESTA

FagusED · Co-ordinated Sciences
C3 · Stoichiometry

B) $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$ — It conserves four H and two O atoms.

COS-0087

RESPUESTA

FagusED · Co-ordinated Sciences
C3 · Stoichiometry

A) 0.25 — M r 44, so $11/44 = 0.25$ mol.

COS-0086

RESPUESTA

FagusED · Co-ordinated Sciences
C3 · Stoichiometry

C) $100 - 40 + 12 + 3 \times 16 = 100$.

COS-0085

RESPUESTA

FagusED · Co-ordinated Sciences
C3 · Stoichiometry

Explain formulae and equations.

COS-0088

PREGUNTA

FagusED · Co-ordinated Sciences
C4 · Electrochemistry

Where does reduction occur during electrolysis?

- A) anode
- B) cathode
- C) electrolyte surface
- D) power supply only

COS-0089

PREGUNTA

FagusED · Co-ordinated Sciences
C4 · Electrochemistry

Why does solid sodium chloride not conduct?

- A) It has no ions.
- B) Its ions are fixed.
- C) Its electrons are delocalised.
- D) Sodium is absent.

COS-0090

PREGUNTA

FagusED · Co-ordinated Sciences
C4 · Electrochemistry

What forms at the cathode during electrolysis of molten PbBr_2 ?

- A) bromine
- B) oxygen
- C) lead
- D) hydrogen

COS-0091

PREGUNTA

FagusED · Co-ordinated Sciences
C4 · Electrochemistry

Explain electrolysis and electrodes.

COS-0092

PREGUNTA

FagusED · Co-ordinated Sciences
C5 · Chemical energetics

Which change is exothermic?

- A) photosynthesis
- B) thermal decomposition
- C) combustion
- D) bond breaking only

COS-0093

PREGUNTA

FagusED · Co-ordinated Sciences
C5 · Chemical energetics

In an endothermic reaction, products have

- A) less energy than reactants
- B) more energy than reactants
- C) zero energy
- D) the same energy always

COS-0094

PREGUNTA

FagusED · Co-ordinated Sciences
C5 · Chemical energetics

Activation energy is the

- A) total energy in products
- B) minimum energy for successful collision
- C) energy released by cooling
- D) temperature change

COS-0095

PREGUNTA

FagusED · Co-ordinated Sciences
C5 · Chemical energetics

Explain energy transfer.

COS-0096

PREGUNTA

FagusED · Co-ordinated Sciences
C6 · Chemical reactions

Which is a chemical change?

- A) melting ice
- B) boiling water
- C) burning magnesium
- D) dissolving sugar

COS-0097

PREGUNTA

FagusED · Co-ordinated Sciences
C6 · Chemical reactions

Rate can be calculated as

- A) time/change
- B) change/time
- C) mass $\times$ time
- D) volume+time

COS-0098

PREGUNTA

FagusED · Co-ordinated Sciences
C6 · Chemical reactions

Why does higher concentration increase rate?

- A) Particles get larger.
- B) More frequent collisions occur.
- C) Activation energy increases.
- D) Bonds vanish.

COS-0099

PREGUNTA

FagusED · Co-ordinated Sciences
C4 · Electrochemistry

B) Its ions are fixed. — The ions cannot carry charge through the solid.

COS-0090

RESPUESTA

FagusED · Co-ordinated Sciences
C4 · Electrochemistry

B) cathode — Cations gain electrons at the cathode.

COS-0089

RESPUESTA

FagusED · Co-ordinated Sciences
C3 · Stoichiometry

A formula shows the simplest whole-number ratio of atoms or ions. Balanced equations conserve the number of each type of atom and therefore mass. State symbols are (s), (l), (g) and (aq). Ionic equations remove spectator ions and show only species that change.

COS-0088

RESPUESTA

FagusED · Co-ordinated Sciences
C5 · Chemical energetics

C) combustion — Combustion transfers energy to surroundings.

COS-0093

RESPUESTA

FagusED · Co-ordinated Sciences
C4 · Electrochemistry

Electrolysis is the decomposition of an ionic compound, molten or in aqueous solution, by electricity. The electrolyte contains mobile ions. The cathode is negative and attracts cations; reduction (gain of electrons) occurs there. The anode is positive and attracts anions; oxidation (loss of electrons) occurs there.

COS-0092

RESPUESTA

FagusED · Co-ordinated Sciences
C4 · Electrochemistry

C) lead — Pb^{2+} ions gain electrons.

COS-0091

RESPUESTA

FagusED · Co-ordinated Sciences
C5 · Chemical energetics

Exothermic reactions transfer thermal energy to the surroundings, causing temperature to rise; products have lower chemical energy than reactants. Endothermic reactions take in thermal energy, causing temperature to fall; products have higher chemical energy. Combustion and neutralisation are exothermic; thermal decomposition and photosynthesis are endothermic.

COS-0096

RESPUESTA

FagusED · Co-ordinated Sciences
C5 · Chemical energetics

B) minimum energy for successful collision — It is the threshold for reaction.

COS-0095

RESPUESTA

FagusED · Co-ordinated Sciences
C5 · Chemical energetics

B) more energy than reactants — The system has absorbed energy.

COS-0094

RESPUESTA

FagusED · Co-ordinated Sciences
C6 · Chemical reactions

B) More frequent collisions occur. — More particles per volume collide more often.

COS-0099

RESPUESTA

FagusED · Co-ordinated Sciences
C6 · Chemical reactions

B) change/time — Rate is change in quantity per unit time.

COS-0098

RESPUESTA

FagusED · Co-ordinated Sciences
C6 · Chemical reactions

C) burning magnesium — Magnesium oxide, a new substance, forms.

COS-0097

RESPUESTA

FagusED · Co-ordinated Sciences
C6 · Chemical reactions

Explain physical and chemical changes.

COS-0100

PREGUNTA

FagusED · Co-ordinated Sciences
C7 · Acids, bases and salts

Which ion causes aqueous acidity?

- A) H⁺
- B) OH⁻
- C) Na⁺
- D) Cl⁻

COS-0101

PREGUNTA

FagusED · Co-ordinated Sciences
C7 · Acids, bases and salts

Acid + carbonate produces salt, water and

- A) hydrogen
- B) oxygen
- C) carbon dioxide
- D) chlorine

COS-0102

PREGUNTA

FagusED · Co-ordinated Sciences
C7 · Acids, bases and salts

Sulfuric acid forms

- A) chlorides
- B) nitrates
- C) sulfates
- D) carbonates

COS-0103

PREGUNTA

FagusED · Co-ordinated Sciences
C7 · Acids, bases and salts

Explain acids, bases and alkalis.

COS-0104

PREGUNTA

FagusED · Co-ordinated Sciences
C8 · The Periodic Table

Elements are arranged by increasing

- A) nucleon number
- B) proton number
- C) density
- D) melting point

COS-0105

PREGUNTA

FagusED · Co-ordinated Sciences
C8 · The Periodic Table

An element with electron arrangement 2,8,2 is in

- A) period 2 group II
- B) period 3 group II
- C) period 2 group III
- D) period 3 group VIII

COS-0106

PREGUNTA

FagusED · Co-ordinated Sciences
C8 · The Periodic Table

Group I metals form ions with charge

- A) 1⁺
- B) 2⁺
- C) 1⁻
- D) 2⁻

COS-0107

PREGUNTA

FagusED · Co-ordinated Sciences
C8 · The Periodic Table

Explain arrangement and electronic structure.

COS-0108

PREGUNTA

FagusED · Co-ordinated Sciences
C9 · Metals

Why is copper used for electrical wiring?

- A) low conductivity
- B) conductivity and ductility
- C) brittleness
- D) high reactivity

COS-0109

PREGUNTA

FagusED · Co-ordinated Sciences
C9 · Metals

Why are alloys often harder than pure metals?

- A) They contain molecules.
- B) Different-sized atoms hinder layer sliding.
- C) They have no electrons.
- D) All atoms become identical.

COS-0110

PREGUNTA

FagusED · Co-ordinated Sciences
C9 · Metals

Which metal reacts most vigorously with cold water?

- A) copper
- B) iron
- C) sodium
- D) silver

COS-0111

PREGUNTA

FagusED · Co-ordinated Sciences
C7 · Acids, bases and salts

C) carbon dioxide — Carbonates release carbon dioxide with acids.

COS-0102

RESPUESTA

FagusED · Co-ordinated Sciences
C7 · Acids, bases and salts

A) H^+ — Acids release hydrogen ions in water.

COS-0101

RESPUESTA

FagusED · Co-ordinated Sciences
C6 · Chemical reactions

A physical change does not form a new substance and is often reversible by physical means. A chemical change rearranges atoms and forms new substances; signs can include gas, precipitate, lasting colour change or temperature change, but evidence should be interpreted in context. Mass is conserved in a closed system.

COS-0100

RESPUESTA

FagusED · Co-ordinated Sciences
C8 · The Periodic Table

B) proton number — Proton number defines the periodic sequence.

COS-0105

RESPUESTA

FagusED · Co-ordinated Sciences
C7 · Acids, bases and salts

Acids produce $H^+(aq)$ ions; bases neutralise acids; alkalis are soluble bases that produce $OH^-(aq)$. Acid + metal gives salt + hydrogen; acid + base gives salt + water; acid + carbonate gives salt + water + carbon dioxide. The salt name comes from the acid: hydrochloric forms chlorides, nitric forms nitrates, sulfuric forms sulfates.

COS-0104

RESPUESTA

FagusED · Co-ordinated Sciences
C7 · Acids, bases and salts

C) sulfates — The acid anion becomes sulfate in the salt.

COS-0103

RESPUESTA

FagusED · Co-ordinated Sciences
C8 · The Periodic Table

Elements are arranged in increasing proton number. A period number gives the number of occupied electron shells; for main-group elements the group number relates to outer-shell electrons. Elements in the same group have similar chemical properties because they have the same number of valence electrons. Across a period, properties change from metallic to non-metallic.

COS-0108

RESPUESTA

FagusED · Co-ordinated Sciences
C8 · The Periodic Table

A) $1+$ — They lose one valence electron.

COS-0107

RESPUESTA

FagusED · Co-ordinated Sciences
C8 · The Periodic Table

B) period 3 group II — Three shells and two outer electrons.

COS-0106

RESPUESTA

FagusED · Co-ordinated Sciences
C9 · Metals

C) sodium — Sodium is high in the reactivity series.

COS-0111

RESPUESTA

FagusED · Co-ordinated Sciences
C9 · Metals

B) Different-sized atoms hinder layer sliding. — Lattice distortion blocks easy movement of layers.

COS-0110

RESPUESTA

FagusED · Co-ordinated Sciences
C9 · Metals

B) conductivity and ductility — It carries charge and can be drawn into wires.

COS-0109

RESPUESTA

FagusED · Co-ordinated Sciences
C9 · Metals

Explain properties and uses.

COS-0112

PREGUNTA

FagusED · Co-ordinated Sciences
C10 · Chemistry of the environment

Potable water is water that is
A) pure H₂O only
B) safe to drink
C) always distilled
D) free of dissolved ions

COS-0113

PREGUNTA

FagusED · Co-ordinated Sciences
C10 · Chemistry of the environment

Anhydrous copper(II) sulfate changes from
A) blue to white
B) white to blue
C) pink to blue
D) green to red

COS-0114

PREGUNTA

FagusED · Co-ordinated Sciences
C10 · Chemistry of the environment

Which element in fertiliser is needed for amino acids?
A) nitrogen
B) argon
C) chlorine
D) neon

COS-0115

PREGUNTA

FagusED · Co-ordinated Sciences
C10 · Chemistry of the environment

Explain water supply and treatment.

COS-0116

PREGUNTA

FagusED · Co-ordinated Sciences
C11 · Organic chemistry

Successive members of a homologous series differ by
A) CH
B) CH₂
C) C₂H₂
D) OH

COS-0117

PREGUNTA

FagusED · Co-ordinated Sciences
C11 · Organic chemistry

Which is an alkane?
A) C₃H₆
B) C₄H₈
C) C₄H₁₀
D) C₂H₄

COS-0118

PREGUNTA

FagusED · Co-ordinated Sciences
C11 · Organic chemistry

Which property increases with hydrocarbon chain length?
A) volatility
B) ease of ignition
C) boiling point
D) flammability

COS-0119

PREGUNTA

FagusED · Co-ordinated Sciences
C11 · Organic chemistry

Explain representing organic compounds.

COS-0120

PREGUNTA

FagusED · Co-ordinated Sciences
C12 · Experimental techniques and chemical...

Which apparatus transfers 25.0 cm³ most accurately?
A) beaker
B) pipette
C) conical flask
D) test tube

COS-0121

PREGUNTA

FagusED · Co-ordinated Sciences
C12 · Experimental techniques and chemical...

Which method separates an insoluble solid from a liquid?
A) filtration
B) distillation
C) chromatography
D) titration

COS-0122

PREGUNTA

FagusED · Co-ordinated Sciences
C12 · Experimental techniques and chemical...

Which method obtains pure water from salt solution?
A) filtration
B) simple distillation
C) crystallisation only
D) paper chromatography

COS-0123

PREGUNTA

B) white to blue — Hydration forms blue copper(II) sulfate.

COS-0114

RESPUESTA

B) safe to drink — It is treated to be safe, though minerals can remain.

COS-0113

RESPUESTA

Metals conduct heat and electricity, are malleable and ductile, and often have high density and melting point. Uses follow properties: copper wiring combines electrical conductivity and ductility; aluminium aircraft parts combine low density, strength and corrosion resistance. Chemical properties vary widely with reactivity.

COS-0112

RESPUESTA

B) CH₂ — Each successive chain has one extra carbon and two hydrogens.

COS-0117

RESPUESTA

Natural water may contain dissolved salts, microorganisms and suspended solids. Potable water is safe to drink, not chemically pure. Treatment commonly includes screening or sedimentation, filtration and chlorination. Distillation produces pure water but requires much energy. Tests for water include anhydrous copper(II) sulfate, white to blue, and cobalt(II) chloride paper, blue to pink.

COS-0116

RESPUESTA

A) nitrogen — Nitrogen is required to make amino acids and proteins.

COS-0115

RESPUESTA

A molecular formula gives numbers of each atom; a displayed formula shows every atom and bond; a structural formula shows connectivity in a compact form. Members of a homologous series share a functional group and general formula, have similar chemical properties, show gradual physical-property trends and differ successively by CH₂.

COS-0120

RESPUESTA

C) boiling point — Stronger intermolecular forces raise boiling point.

COS-0119

RESPUESTA

C) C₄H₁₀ — Alkanes follow C_nH_{2n+2}.

COS-0118

RESPUESTA

B) simple distillation — Water vaporises and is condensed.

COS-0123

RESPUESTA

A) filtration — The solid remains in the filter.

COS-0122

RESPUESTA

B) pipette — A volumetric pipette transfers a fixed accurate volume.

COS-0121

RESPUESTA

Explain measurement and apparatus.

COS-0124

PREGUNTA

How is average speed calculated?

COS-0125

PREGUNTA

How is acceleration calculated?

COS-0126

PREGUNTA

What does the gradient of a distance-time graph represent?

COS-0127

PREGUNTA

What does the gradient of a velocity-time graph represent?

COS-0128

PREGUNTA

How does a gas create pressure on a container wall?

COS-0129

PREGUNTA

Why does pressure increase when a sealed gas is heated at constant volume?

COS-0130

PREGUNTA

What is internal energy?

COS-0131

PREGUNTA

What is specific heat capacity?

COS-0132

PREGUNTA

Which quantity is measured in hertz?
A) frequency
B) wavelength
C) amplitude
D) wave speed

COS-0133

PREGUNTA

A wave has frequency 4.0 Hz and wavelength 0.50 m. What is its speed?
A) 0.125 m/s
B) 2.0 m/s
C) 4.5 m/s
D) 8.0 m/s

COS-0134

PREGUNTA

Which wave is longitudinal?
A) light in vacuum
B) sound in air
C) radio wave
D) wave on a string

COS-0135

PREGUNTA

Explain wave quantities.

COS-0136

PREGUNTA

Which equation defines current?
A) $I = Q/t$
B) $I = Qt$
C) $I = VQ$
D) $I = E/t$

COS-0137

PREGUNTA

A 6.0 V resistor carries 0.50 A. Its resistance is
A) 3.0 Ω
B) 12 Ω
C) 30 Ω
D) 0.083 Ω

COS-0138

PREGUNTA

FagusED · Co-ordinated Sciences
P1 · Motion, forces and energy

Acceleration = change in velocity ÷ time taken; it has units m/s².

COS-0126

RESPUESTA

FagusED · Co-ordinated Sciences
P1 · Motion, forces and energy

Average speed = total distance ÷ total time; include a suitable speed unit.

COS-0125

RESPUESTA

FagusED · Co-ordinated Sciences
C12 · Experimental techniques and chemical...

Choose apparatus according to precision and purpose. A burette measures variable accurate volumes; a volumetric pipette transfers one fixed accurate volume; a measuring cylinder is less precise; a gas syringe measures gas volume. Read liquid levels at eye level from the bottom of a concave meniscus and record burette readings to the appropriate precision.

COS-0124

RESPUESTA

FagusED · Co-ordinated Sciences
P2 · Thermal physics

Gas particles collide with the wall, changing momentum and exerting force over its area.

COS-0129

RESPUESTA

FagusED · Co-ordinated Sciences
P1 · Motion, forces and energy

Acceleration; the signed area between the graph and time axis gives displacement.

COS-0128

RESPUESTA

FagusED · Co-ordinated Sciences
P1 · Motion, forces and energy

Speed. A steeper line means a greater speed.

COS-0127

RESPUESTA

FagusED · Co-ordinated Sciences
P2 · Thermal physics

Energy required to raise the temperature of 1 kg of a substance by 1 °C; $E = mc\Delta T$.

COS-0132

RESPUESTA

FagusED · Co-ordinated Sciences
P2 · Thermal physics

The sum of particles' random kinetic energies and their potential energies from particle interactions.

COS-0131

RESPUESTA

FagusED · Co-ordinated Sciences
P2 · Thermal physics

Particles gain kinetic energy; collisions with the walls become more frequent and cause larger momentum changes.

COS-0130

RESPUESTA

FagusED · Co-ordinated Sciences
P3 · Waves

B) sound in air

COS-0135

RESPUESTA

FagusED · Co-ordinated Sciences
P3 · Waves

B) 2.0 m/s

COS-0134

RESPUESTA

FagusED · Co-ordinated Sciences
P3 · Waves

A) frequency

COS-0133

RESPUESTA

FagusED · Co-ordinated Sciences
P4 · Electricity and magnetism

B) 12 Ω

COS-0138

RESPUESTA

FagusED · Co-ordinated Sciences
P4 · Electricity and magnetism

A) $I = Q/t$

COS-0137

RESPUESTA

FagusED · Co-ordinated Sciences
P3 · Waves

A wave transfers energy without a net transfer of matter. Amplitude is maximum displacement from equilibrium; wavelength is distance between corresponding points; frequency is oscillations per second; period is time for one oscillation.

COS-0136

RESPUESTA

FagusED · Co-ordinated Sciences
P4 · Electricity and magnetism

What is the current at every point in a series circuit?

- A) always zero
- B) the same
- C) different at each resistor
- D) equal to voltage

COS-0139

PREGUNTA

FagusED · Co-ordinated Sciences
P4 · Electricity and magnetism

Explain charge, current and potential difference.

COS-0140

PREGUNTA

FagusED · Co-ordinated Sciences
P5 · Nuclear physics

Which particles are in the nucleus?

- A) protons and neutrons
- B) protons and electrons
- C) neutrons and electrons
- D) electrons only

COS-0141

PREGUNTA

FagusED · Co-ordinated Sciences
P5 · Nuclear physics

Two isotopes of the same element have different numbers of

- A) protons
- B) electrons in neutral atoms
- C) neutrons
- D) nuclear charge

COS-0142

PREGUNTA

FagusED · Co-ordinated Sciences
P5 · Nuclear physics

What is random about radioactive decay?

- A) the type of element
- B) which nucleus decays and when
- C) the existence of a nucleus
- D) the charge of a proton

COS-0143

PREGUNTA

FagusED · Co-ordinated Sciences
P5 · Nuclear physics

Explain atomic structure and nuclide notation.

COS-0144

PREGUNTA

FagusED · Co-ordinated Sciences
P6 · Space physics

What provides the centripetal force for a planet orbiting the Sun?

- A) gravity
- B) magnetism
- C) air resistance
- D) nuclear force

COS-0145

PREGUNTA

FagusED · Co-ordinated Sciences
P6 · Space physics

A planet completes an orbit of radius r in period T . Its orbital speed is

- A) r/T
- B) $2\pi r/T$
- C) $T/2\pi r$
- D) $2\pi T/r$

COS-0146

PREGUNTA

FagusED · Co-ordinated Sciences
P6 · Space physics

Why is an orbiting satellite accelerating at constant speed?

- A) its mass changes
- B) its direction changes
- C) gravity is zero
- D) its period is infinite

COS-0147

PREGUNTA

FagusED · Co-ordinated Sciences
P6 · Space physics

Explain the solar system and gravity.

COS-0148

PREGUNTA

FagusED · Co-ordinated Sciences
P5 · Nuclear physics

A) protons and neutrons

COS-0141

RESPUESTA

FagusED · Co-ordinated Sciences
P4 · Electricity and magnetism

Electric charge is positive or negative. Current is the rate of flow of charge; conventional current is from positive to negative outside a source. Potential difference is energy transferred per unit charge, while e.m.f. is energy supplied per unit charge by a source.

COS-0140

RESPUESTA

FagusED · Co-ordinated Sciences
P4 · Electricity and magnetism

B) the same

COS-0139

RESPUESTA

FagusED · Co-ordinated Sciences
P5 · Nuclear physics

An atom has a tiny positive nucleus containing protons and neutrons, surrounded by electrons. Proton number Z identifies the element; nucleon number A is protons plus neutrons. Isotopes have the same Z but different neutron numbers.

COS-0144

RESPUESTA

FagusED · Co-ordinated Sciences
P5 · Nuclear physics

B) which nucleus decays and when

COS-0143

RESPUESTA

FagusED · Co-ordinated Sciences
P5 · Nuclear physics

C) neutrons

COS-0142

RESPUESTA

FagusED · Co-ordinated Sciences
P6 · Space physics

B) its direction changes

COS-0147

RESPUESTA

FagusED · Co-ordinated Sciences
P6 · Space physics

B) $2\pi r/T$

COS-0146

RESPUESTA

FagusED · Co-ordinated Sciences
P6 · Space physics

A) gravity

COS-0145

RESPUESTA

FagusED · Co-ordinated Sciences
P6 · Space physics

The Solar System contains the Sun, eight planets, dwarf planets, moons, asteroids and comets. Gravity provides the centripetal force for orbits. Orbital speed depends on orbital radius and period; closer planets generally move faster.

COS-0148

RESPUESTA